

# Algorithmic Construction and Exploitation of Topology-Constrained Refinement Models for Semiexperimental Equilibrium Structures

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Semiexperimental equilibrium structures combine high-resolution rotational spectroscopy with computed vibrational and electronic corrections. The least-squares solution of a chosen refinement model is now routine; the less formalized step is the construction of that model. Active coordinates, symmetry, weakly determined directions, constraints, and parameter classes are still often selected by expert intervention, especially for cyclic, fused, and bridged molecular graphs.

This paper introduces MORPHEUS (Model Optimization and Refinement from Perceived Hierarchy, Equivalence, Uncertainty, and Symmetry), a framework that turns model formulation into an auditable topology-perception problem. The semiexperimental refinements reported here are the principal validation, but the construction is more general: it provides an algorithmic way to build constrained refinement models from topology, symmetry, rank, and external reference information before any numerical least-squares solution is attempted. The resulting models are not merely topology-aware: the active representation and all constraint or parameter-class relations are generated from the perceived graph, symmetry-consistent, rank-consistent, and recorded. Molecular topology defines the primitive coordinate space, symmetry selects the admissible active subspace, isotopic coverage determines which directions are supported, and electronic-structure reference information enters only through explicit constraints, predicates, or parameter classes.

Two equivalent active spaces are available: symmetry-adapted generalized internal coordinates built from homogeneous primitive families, and symmetry-adapted Cartesian displacements after removal of translations and rotations. The chemical interface remains internal-coordinate based: constraints, predicates, and parameter classes are expressed as primitive bond, angle, dihedral, out-of-plane, ring, or local-frame relations and then projected onto the selected active basis. Validation on acyclic, cyclic, planar aromatic, fused, bridged, and flexible benchmarks shows that the same rank, effective dimension, and rotational residuals are obtained without manual coordinate-model formulation, while final structural parameters and uncertainties remain chemically interpretable. A 49-atom testosterone scale check further shows that quantum chemical equilibrium geometries, accelerated vibrational corrections, and class-constrained model generation can be combined for large molecular systems.

**Keywords:** semiexperimental equilibrium structures, rotational spectroscopy, generalized internal coordinates, least squares, Kraitchman equations

## I. INTRODUCTION

Accurate equilibrium molecular structures are central to rotational spectroscopy, thermochemistry, force-field development, and the validation of electronic-structure methods.<sup>1,2</sup> Despite decades of progress in semiexperimental refinement, model construction remains largely a manual activity. Structure determination from experimental observables is also a coordinate-dependent inverse problem: the data constrain a Cartesian geometry, but the fitted variables determine conditioning, constraints, correlations, and uncertainties. Rotational spectroscopy is a stringent example because rotational constants probe the molecular mass distribution with high precision, whereas semiexperimental ( $r_e^{\text{SE}}$ ) structures require vibrational and sometimes electronic corrections to convert ground-state constants into equilibrium information.<sup>3–6</sup>

Modern fitting programs and workflows<sup>7–9</sup> have estab-

lished robust statistical principles: fit well-conditioned observables, include theoretical predicates when experimental information is incomplete, and propagate covariance to derived structural parameters.<sup>10</sup> Their common prerequisite is that the refinement model has already been formulated. That prerequisite is not innocent, especially for rings, fused systems, and bridged frameworks. A molecular graph with cycles is not a tree; different reference paths can make one distance, angle, or torsion primary and an equivalent one derived. For such systems, construction of a symmetry-consistent non-redundant refinement model is itself a major part of the semiexperimental analysis. The target of the present work is therefore not a new least-squares observable or a molecule-specific improvement in residuals, but an algorithmic route from perceived topology to a reproducible statistical refinement model. Although semiexperimental structure refinement is the principal use case developed here, the same construction applies more generally to constrained refinement models in which topology, symmetry, rank, and external information must be combined before the numerical fit is solved.

Recent workflows have extended semiexperimental re-

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finement by combining computed starting geometries, quantum chemical (QC) vibrational corrections, and reduced-dimensionality models.<sup>11</sup> The implementation developed here uses such data as upstream inputs while making the refinement model itself graph-defined rather than hand specified. Reference geometries from QC computations or from fragment libraries (e.g., LCB25<sup>12</sup>) enter only as chemically explicit constraints, predicates, or parameter classes; the fitted variables remain the algorithmically generated non-redundant active coordinates. This separation is important for larger molecules. Accurate QC equilibrium geometries and vibrational corrections are prerequisites for comparison with ground-state rotational constants; the framework adds the missing step of converting those ingredients into an explicit, rank-audited refinement model without molecule-specific coordinate design.

The procedure starts from Cartesian coordinates. Primitive GICs are built, redundancies are removed within homogeneous coordinate families, symmetry is assigned, and the totally symmetric active space is defined. Isotope coverage, symmetry orbits, chemical descriptors, and reference data then generate compatible constraints, predicates, and parameter classes. Two active spaces are available: symmetry-adapted GICs and symmetry-adapted Cartesians after removal of translations and rotations. Constraints remain primitive internal-coordinate relations and are projected onto either active basis by the Wilson matrix or the corresponding derivative operator.<sup>13</sup>

The rotational semiexperimental problem is used here as a demanding test because it exposes both rank deficiencies and chemically meaningful constraints. Glycolaldehyde is retained as a standard acyclic case. Cyclopentadiene, nitrobenzene, azulene, and norcamphor test ring closures, planar-component selection, fused topology, bridged topology, and hydrogen-frame constraints. The maleic, phthalic, and succinic anhydrides form a homologous literature-data series with different degrees of isotopic completeness and different auxiliary relations. Two glycine conformers test reduced-dimensionality constraints, and testosterone closes the study as a large-molecule scale demonstration.

## II. THEORY AND METHOD

### A. Overview of the Framework

The overall architecture is summarized in Figure 1. The key separation is between the chemical model, the numerical active basis, and the least-squares solution. Chemical information is always expressed in primitive internal coordinates; the active variables may be GIC amplitudes or symmetry-adapted Cartesian displacements. This separation is what makes the generated model auditable: the same primitive relation has the same chemical meaning regardless of the basis used to solve the

refinement.

### B. Primitive and Work Coordinates

The coordinate construction starts from Cartesian geometry and perceived topology. Bonded pairs and ring connectivity define a primitive set of internal coordinates: stretches, valence angles, torsions, out-of-plane deformations, linear bends, and ring-specific primitives. Redundancy is intentional at this stage because it lets topology, not a manually chosen reference path, decide which combinations survive. For cyclic systems, endocyclic valence-angle combinations describe ring bending, whereas the endocyclic torsional manifold is reported through ring-puckering descriptors such as  $q$  and  $\phi$ . These descriptors are constructed from the underlying dihedral combinations but are more stable and interpretable than individual endocyclic torsions, so monocyclic, fused-ring, bridged, and polycyclic systems are handled by the same machinery.<sup>14</sup> This primitive layer is also the chemical interface for constraints, predicates, and parameter classes.

### C. Coordinates for Parameter Refinement

The active space must be non-redundant, symmetry-consistent, and compatible with chemically meaningful constraints. In the GIC route, topological primitives are reduced within homogeneous families, so stretches, bends, torsions, out-of-plane motions, and ring coordinates cannot replace one another during rank reduction.<sup>15–18</sup> In the Cartesian route, translations and rotations are removed from the Cartesian displacement space. In both cases symmetry selects the totally symmetric active block and completes primitive constraints over equivalent orbits.

At a reference geometry  $\mathbf{x}_0$ , primitive displacements and Cartesian displacements are related to first order by the Wilson matrix<sup>13</sup>

$$\Delta \mathbf{s} = \mathbf{B}_s \Delta \mathbf{x}, \quad \mathbf{B}_s = \left. \frac{\partial \mathbf{s}}{\partial \mathbf{x}} \right|_{\mathbf{x}_0}. \quad (1)$$

For each homogeneous primitive family  $\alpha$ , the Wilson block  $\mathbf{B}_{s_\alpha}$  is submitted to a singular-value decomposition,<sup>19</sup>

$$\mathbf{B}_{s_\alpha} = \frac{\partial \mathbf{s}_\alpha}{\partial \mathbf{x}}. \quad (2)$$

$$\mathbf{B}_{s_\alpha} = \mathbf{U}_\alpha \Sigma_\alpha \mathbf{V}_\alpha^T, \quad (3)$$

The retained left-singular vectors define the row-orthonormal transformation

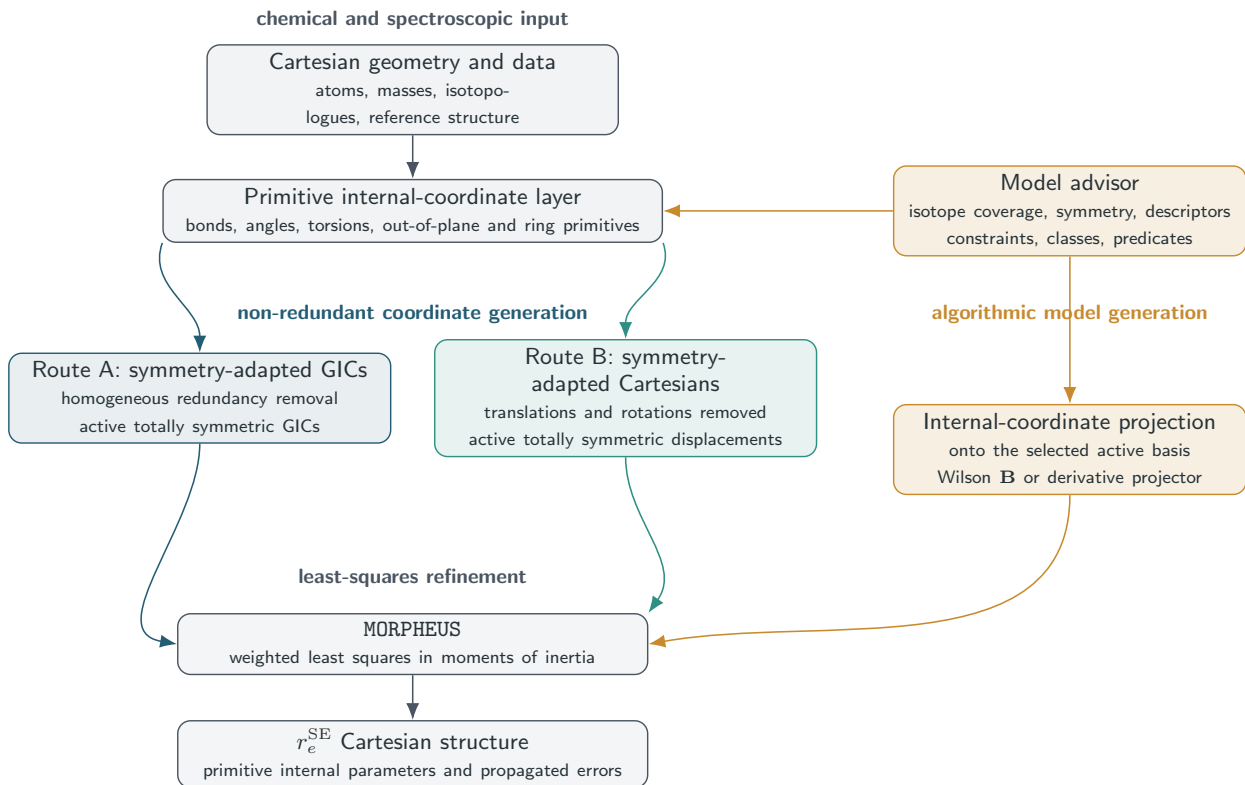


FIG. 1. Architecture of the model-generation framework.

$$\mathbf{R}_\alpha = \mathbf{U}_{\alpha,r}^T, \quad (4)$$

where the subscript  $r$  denotes the retained singular subspace. The non-redundant coordinates for family  $\alpha$  are then

$$\Delta \mathbf{q}_\alpha = \mathbf{R}_\alpha \Delta \mathbf{s}_\alpha, \quad \mathbf{B}_{q_\alpha} = \mathbf{R}_\alpha \mathbf{B}_{s_\alpha}. \quad (5)$$

The retained rank in each family gives the number of coordinates in that family. Stacking the blocks gives

$$\Delta \mathbf{q} = \mathbf{R} \Delta \mathbf{s}, \quad \mathbf{B}_q = \mathbf{R} \mathbf{B}_s. \quad (6)$$

The block-diagonal structure preserves chemical homogeneity. A stretch can therefore not remove a torsion from the model, and a ring coordinate cannot be lost because a different primitive family happened to enter the rank test first. After family-wise reduction, residual dependencies are pruned with deterministic topological tie breaking and the final basis is checked against the expected vibrational rank. This is a restriction on the admissible coordinate transformations, not on the rank test itself. A global SVD of the full redundant primitive set would also produce an orthonormal basis for the row space, but its left singular vectors are free to mix primitives with different chemical meaning whenever their

Cartesian derivatives are linearly coupled or nearly degenerate. The numerical rank would be correct, whereas the resulting coordinates could contain arbitrary stretch-bend-torsion admixtures whose composition changes with scaling, ordering, or small geometry perturbations. Performing the rank reduction in homogeneous primitive families instead preserves the direct map between each retained coordinate and the chemical relation used for constraints, predicates, classes, and uncertainty propagation; only residual linear dependencies between already interpretable coordinates are removed in the final pruning step. Cartesian displacements associated with GIC corrections use a rank-controlled minimum-norm pseudoinverse with explicit singular-value thresholds. The symmetry-Cartesian route uses the same selection, constraint projection, and covariance propagation.

#### D. Constraints, Predicates, and Parameter Classes

Structural relations are specified in primitive internal coordinates,

$$\mathbf{s} = (s_1, s_2, \dots, s_m)^T, \quad (7)$$

and may be supplied by the user or generated by the advisor. The same projector is used for constraints, predicates, classes, and error propagation, so changing from

GICs to symmetry-adapted Cartesians changes the numerical basis, not the primitive chemical relations that define the model. Primitive-coordinate predicates are parsed canonically, and reversible angle or torsion orderings are normalized before projection.

*a. Hard constraints.* Hard constraints are exact equations imposed throughout the refinement:

$$s_i = s_i^0, \quad (8)$$

equalities,

$$s_i - s_j = 0, \quad (9)$$

or general linearized functions,

$$\mathbf{c}(\mathbf{s}) = \mathbf{d}. \quad (10)$$

At a trial geometry the corresponding first-order condition is

$$\mathbf{C}_s \Delta \mathbf{s} = \mathbf{b}_c, \quad \mathbf{C}_s = \left. \frac{\partial \mathbf{c}}{\partial \mathbf{s}} \right|_{\mathbf{s}_0}, \quad \mathbf{b}_c = \mathbf{d} - \mathbf{c}(\mathbf{s}_0). \quad (11)$$

Projection to the active space gives

$$\mathbf{C}_A \Delta \mathbf{a} = \mathbf{b}_c, \quad \mathbf{C}_A = \mathbf{C}_s \mathbf{B}_s \mathbf{T}_A. \quad (12)$$

The solver keeps all compatible motions by writing the allowed correction as

$$\Delta \mathbf{a} = \Delta \mathbf{a}_0 + \mathbf{N} \Delta \mathbf{p}, \quad \mathbf{C}_A \mathbf{N} = \mathbf{0}, \quad (13)$$

where  $\Delta \mathbf{a}_0$  is a minimum-norm particular solution and  $\mathbf{N}$  spans the null space. The least-squares problem is solved only for  $\Delta \mathbf{p}$ .

Symmetry-equivalent primitives are constrained as complete orbits before the null-space projector is formed. Reduced-dimensionality models use the same projection, allowing unsupported hydrogen frames to be frozen or tied without altering the covalent topology used for GIC generation.

*b. Predicate observations.* Predicates are soft structural information: they add weighted residuals rather than removing degrees of freedom.<sup>10</sup> A primitive predicate has the form

$$f(\mathbf{s}) = f^{\text{ref}} \pm \sigma_f, \quad (14)$$

and contributes

$$r_f = \frac{f(\mathbf{s}) - f^{\text{ref}}}{\sigma_f} \quad (15)$$

to the residual vector. Its Jacobian with respect to the independent least-squares variables is

$$\mathbf{J}_f = \frac{1}{\sigma_f} \frac{\partial f}{\partial \mathbf{s}} \mathbf{B}_s \mathbf{T}_A \mathbf{N}. \quad (16)$$

*c. Parameter classes.* Parameter classes make several primitive quantities share one correction. If class  $c$  contains primitives  $i \in c$ , the allowed correction is

$$\Delta s_i = \Delta p_c, \quad i \in c. \quad (17)$$

Class definitions are converted into linear relations or Jacobian compression. They apply to corrections rather than final values, so related quantities may retain different reference values while sharing a fitted offset. After convergence, the covariance is expanded back to primitive and topological parameters.

## E. Semiexperimental Observables

For each isotopologue  $s$ , the measured ground-state rotational constants are converted into equilibrium rotational constants by removing vibrational and, when necessary, electronic contributions,<sup>3-6</sup>

$$B_{\alpha,e}^{(s)} = B_{\alpha,0}^{(s)} - \Delta B_{\alpha,\text{vib}}^{(s)} - \Delta B_{\alpha,\text{elec}}^{(s)}, \quad \alpha \in \{a, b, c\}. \quad (18)$$

The correction terms may come from any electronic-structure or composite protocol and are supplied with uncertainties, masses, conventions, and provenance.<sup>11</sup>

Although rotational constants may be fitted directly, the refinement uses principal moments of inertia as default observables, because they give a smoother geometry dependence and avoid reciprocal amplification:

$$I_{\alpha}^{(s)} = \frac{K}{B_{\alpha,e}^{(s)}}, \quad (19)$$

where  $K$  is the standard rotational conversion constant.

For planar molecules only two principal components are independent. If the molecular plane is the  $ab$  plane,

$$I_c = I_a + I_b, \quad (20)$$

Fitting all three components would duplicate one constraint. The workflow therefore uses one pair,  $AB$ ,  $AC$ , or  $BC$ , and reports the omitted component as a diagnostic. If the pair is not specified, all admissible pairs are tested; the selected pair maximizes rank and minimizes the propagated instability of the omitted moment.

The selected observables and uncertainties for all isotopologues form the experimental part of the least-squares objective.

## F. Least-Squares Refinement

As usual,<sup>1,7</sup> the semiexperimental refinement is formulated as a weighted nonlinear least-squares problem. Af-

ter hard constraints and parameter classes have been projected, the actual independent least-squares variables are collected in  $\mathbf{p}$ .

The residual vector is

$$\mathbf{r} = \mathbf{y}^{\text{obs}} - \mathbf{y}^{\text{calc}}(\mathbf{p}), \quad (21)$$

where  $\mathbf{y}$  collects all experimental observables and predicate observations included in the refinement.

The weighted objective is

$$\chi^2 = \mathbf{r}^T \mathbf{W} \mathbf{r}, \quad (22)$$

where  $\mathbf{W}$  contains the inverse variances of experimental and predicate observations. Robust fits reweight complete isotopologue blocks rather than individual rotational constants, while predicates retain their assigned statistical weights.

The optimization is performed using a weighted Levenberg–Marquardt procedure,

$$(\mathbf{J}^T \mathbf{W} \mathbf{J} + \lambda \mathbf{I}) \Delta \mathbf{p} = \mathbf{J}^T \mathbf{W} \mathbf{r}, \quad (23)$$

where  $\mathbf{J}$  is the Jacobian matrix of the observables with respect to the independent variables  $\mathbf{p}$  and  $\lambda$  is the damping parameter. Accepted steps are checked against the original topology, point group, irreducible-representation counts, and coordinate-family counts; topology-changing steps are rejected.

Convergence requires both parameter corrections and objective reduction to fall below thresholds. Diagnostic leave-one-isotopologue-out refits, small singular values, reduced rank, downweighted isotopologues, unstable planar pairs, and large propagated coordinate uncertainties are reported without redefining the coordinate model.

Following convergence, the covariance matrix in the independent fitted variables is obtained from

$$\mathbf{C}_p = \sigma_r^2 (\mathbf{J}^T \mathbf{W} \mathbf{J})^{-1}, \quad (24)$$

where  $\sigma_r^2$  is the reduced residual variance.

### G. Error Propagation

Uncertainty propagation follows the geometry-update chain. If  $\mathbf{N}$  is the null-space basis of the hard constraints and class reductions, the covariance in the active coordinate space is

$$\mathbf{C}_a = \mathbf{N} \mathbf{C}_p \mathbf{N}^T. \quad (25)$$

The corresponding Cartesian covariance is

$$\mathbf{C}_x = \mathbf{T}_A \mathbf{C}_a \mathbf{T}_A^T. \quad (26)$$

For a GIC refinement,  $\mathbf{T}_A = \mathbf{B}_Q^+$ ; for a Cartesian refinement,  $\mathbf{T}_A = \mathbf{C}_{A_1}$ . Thus both coordinate choices use the same covariance propagation.

Reported structural quantities are functions of the optimized Cartesian geometry. For a vector of quantities  $\mathbf{g}(\mathbf{x})$ , which may contain bond lengths, valence angles, dihedral angles, out-of-plane coordinates, ring descriptors, or user-defined primitive functions, the covariance is

$$\mathbf{C}_g = \mathbf{G}_x \mathbf{C}_x \mathbf{G}_x^T, \quad \mathbf{G}_x = \left. \frac{\partial \mathbf{g}}{\partial \mathbf{x}} \right|_{\mathbf{x}_{\text{opt}}}. \quad (27)$$

Standard errors and correlations are obtained from  $\mathbf{C}_g$ . Hard constraints have no variance along removed directions, whereas predicate uncertainties enter the weights and contribute to  $\mathbf{C}_p$ .

The report lists propagated structural errors, correlations, SVD diagnostics of the final weighted Jacobian, projected constraints and classes, and matched active-coordinate labels. Final bond lengths, angles, dihedrals, and ring parameters are always evaluated from the optimized Cartesian geometry, so the reported tables are independent of whether the fit used GIC or Cartesian amplitudes. A numerical Hessian in the final independent variable space gives the stationary-point diagnostic.

### H. Kraitchman Diagnostic

For singly substituted isotopologues, the workflow computes substitution coordinates from the Kraitchman equations<sup>20</sup> and compares their absolute values with the fitted coordinates in the parent principal-axis frame. Kraitchman coordinates are widely used in spectroscopic structure work, either directly as  $r_s$  substitution coordinates, as checks on  $r_0$  or semiexperimental structures, or as practical atom-assignment diagnostics. In this workflow they are intentionally kept outside the least-squares objective: they are useful checks, but they do not retain coordinate signs and do not exploit the full correlated model. At the same experimental-information level and computational cost, the SE+fit route is therefore more informative because all substituted constants enter one constrained Cartesian fit, uncertainties are propagated through the same model, and the omitted planar component can remain an external diagnostic.

## III. IMPLEMENTATION AND REPRODUCIBILITY

The implementation is organized around two reusable services. The coordinate-definition service builds the molecular coordinate model from Cartesian atoms and nuclear charges, assigning point-group symmetry and irreducible representations when requested. A separate B-matrix service evaluates the same frozen coordinate defi-

dition for any compatible Cartesian geometry. During refinement, the engine reuses the stored coordinate model and updates only the derivative information needed for the current step.

The standard input contains the Cartesian parent geometry, optional primitive freeze or constraint records, and isotopologue data with rotational constants, corrections, uncertainties, masses, and conventions.

Each run records the final Cartesian structure, propagated primitive parameters, rotational residuals, covariance and SVD diagnostics, projected constraints/classes, provenance, warnings, and minimum-check Hessian eigenvalues. These outputs provide the audit trail for the algorithmically generated model and are the numerical source for the benchmark tables.

#### IV. RELATIONSHIP TO EXISTING APPROACHES

The framework combines established semiexperimental least-squares machinery<sup>7,10,11</sup> with an algorithmic model-generation layer. The statistical model uses moments of inertia, vibrational and electronic corrections, predicates, parameter classes, leverage and correlation diagnostics, robust weighting, and final minimum checks. The new element is the formal definition and audit trail of the refinement variables, constraints, and parameter-class structure. Existing approaches automate the solution of a model; this work formalizes the generation of the model itself, and then solves it with the same statistical discipline.

Z-matrices remain compact for tree-like molecules, as illustrated below by glycolaldehyde. Their limitations become visible for rings, fused systems, and bridged frameworks: a molecular graph with cycles cannot be represented by a single reference tree without making some chemically equivalent quantities primary and others derived. In a least-squares refinement this affects not only notation but also the covariance attached to reported parameters.

Generalized internal coordinates are a flexible alternative and are widely used in optimization and vibrational analysis.<sup>13,15–18,21</sup> In those workflows the generalized coordinates are usually a user-defined layer. Here they are generated from topology-based primitive coordinates, reduced to a non-redundant space, assigned to irreducible representations, and accessed through primitive-coordinate constraints. Kisiel’s STRFIT/PROSPE route<sup>8</sup> provides another important reference point for rotational-structure fitting, but the structural model is again supplied by the user and may leave derived ring quantities outside the primary fitted covariance matrix.<sup>9</sup>

The practical distinction is summarized in Table I. The workflow retains the usual semiexperimental inputs and diagnostics while moving coordinate construction, symmetry completion, constraint and class generation, and covariance propagation to a common graph, symmetry, and rank layer. The user still controls the chemical prob-

lem and the quality of the input data; what is removed is the need to hand-build the non-redundant refinement model.

#### A. Computational Reference Data

The electronic-structure quantities used in this work were generated with the hierarchical family of Pisa composite schemes (PCSs). In most cases, vibrational corrections and constraint targets were taken from the double-hybrid member of the family (DPCS3), which has been validated in previous work for accurate structures and fast rovibrational corrections. The only exception is conformer I of glycine, where the best available reference was used: PCS4, based on explicitly correlated CCSD(T)-F12 calculations close to the complete-basis-set limit with an additive core–valence correction. Outside this role as reference information, the electronic-structure data do not define the fitted coordinates; they enter only through explicit constraints, predicates, parameter classes, and  $\Delta B_{\text{vib}}$  corrections.

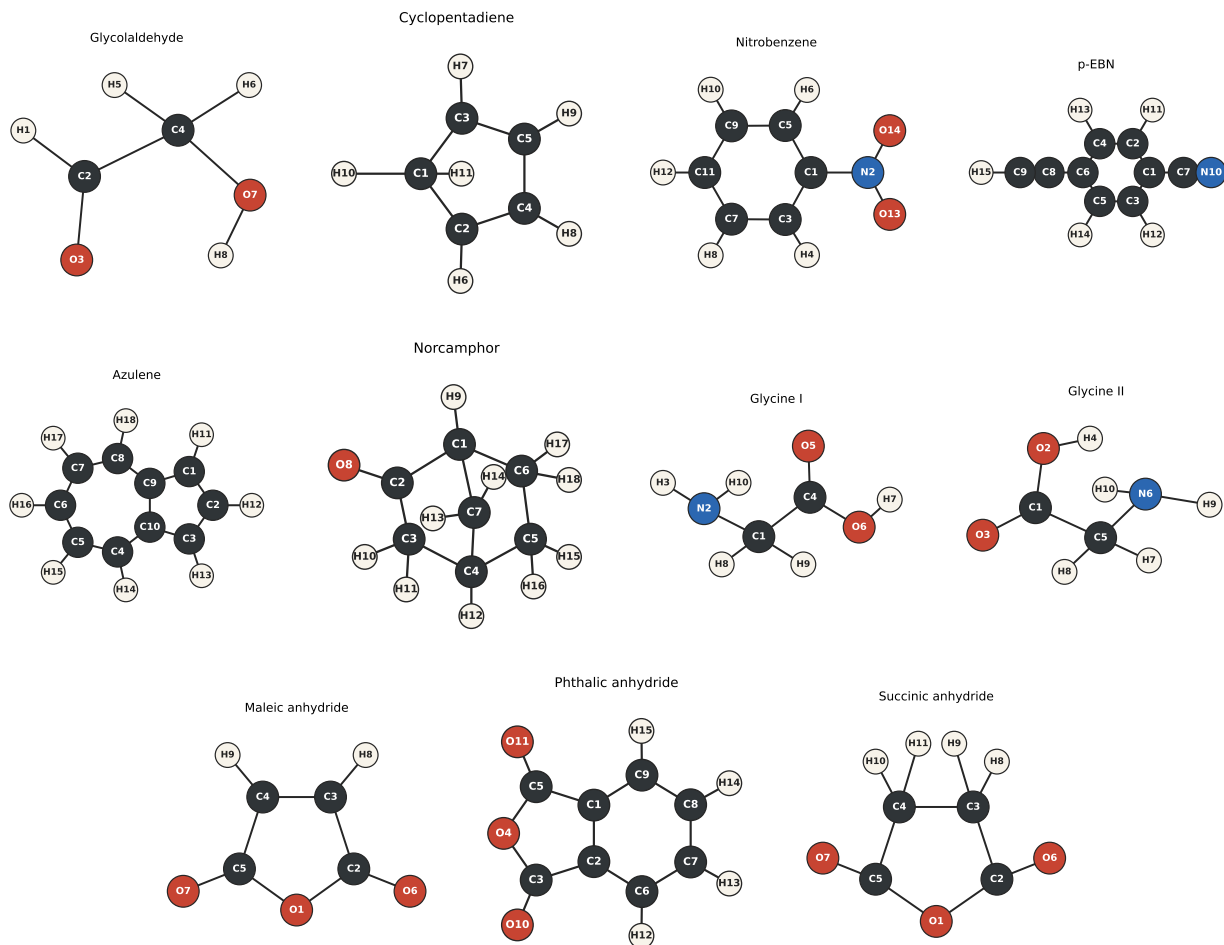
#### V. RESULTS AND DISCUSSION

The systems are discussed in the order in which they test increasingly demanding parts of the same model-generation problem. Glycolaldehyde is the acyclic reference case; cyclopentadiene tests ring closure; nitrobenzene and azulene test planar-component selection and fused aromatic topology; norcamphor tests bridged graphs and hydrogen-frame constraints. The anhydride series then probes how published data sets with different isotopic completeness are converted into equivalent input models. Glycine I and II test reduced-dimensionality constraints. Testosterone is kept separate and last because it is a scale demonstration rather than an isotope-rich benchmark. In all cases the input is Cartesian and no molecule-specific coordinate code, manual reference tree, or manual active-space selection is used.

Figure 3 is the central validation of the method: it quantifies how topology perception, symmetry analysis, isotopic coverage, and reference information are transformed into the final refinement model. The figure is therefore a model-generation audit, not simply a residual summary: it reports how many primitive/non-redundant coordinates are produced, how many remain active after symmetry selection, how many survive constraint projection, and how many constraints are introduced without manual parameter selection. The rotational residuals remain in the expected semiexperimental range for standard, cyclic, planar, fused, bridged, and glycine benchmarks, but the central result is that these residuals are obtained from generated, auditable models. The glycine examples also separate two roles of theory: Gly-I starts from a reference that already satisfies the practical rotational-accuracy target, whereas Gly-

TABLE I. Compact operational comparison of representative structure-refinement routes. The emphasis is model formulation rather than the statistical quality of a particular fit.

| Feature          | Z-matrix           | User GIC or PROSPE | MORPHEUS                 |
|------------------|--------------------|--------------------|--------------------------|
| Input model      | Internal tree      | User model         | Cartesian topology       |
| Active variables | Manual             | Manual             | Rank-pruned              |
| Rings/bridges    | Tree-dependent     | User-defined       | Graph-derived primitives |
| Missing isotopes | Manual constraints | Manual classes     | Generated constraints    |
| Symmetry         | In model           | User supplied      | Detected and audited     |
| Output audit     | Supplied model     | Workflow dependent | Manifest and covariance  |



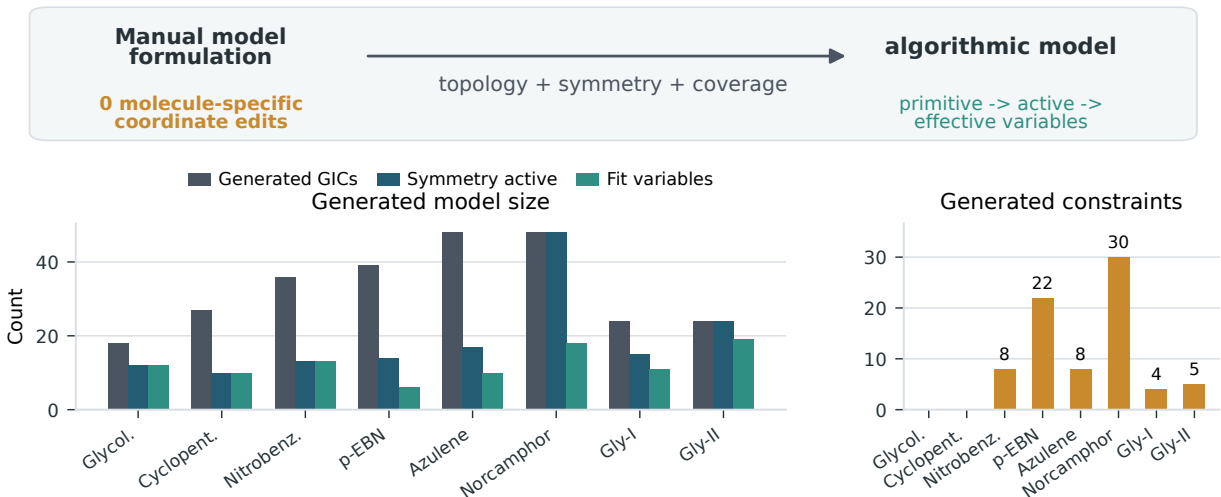


FIG. 3. Algorithmic model-generation audit for the representative refinements. The top guide links the manual formulation problem to the generated model: topology perception, symmetry analysis, and isotopic coverage replace molecule-specific coordinate edits. The left panel reports the number of coordinates generated algorithmically, the symmetry-active subset before constraint projection, and the final least-squares variables after reduction. The right panel reports the number of constraints or frozen primitive parameters projected by the model layer. Manual coordinate-model edits are zero for every case. Detailed fit residuals and ranks are collected in the Supporting Information.

unsupported directions require constraints, which planar rotational components are admissible, and which parameters must remain active after rank projection. In a manual workflow these choices precede the least-squares solution and are a common source of non-reproducibility. In the reported runs they are made by topology perception, symmetry assignment, coverage analysis, and rank checks. Stretches, bends, torsions, out-of-plane terms, linear bends, and ring-deformation coordinates are generated from the perceived graph and reduced in homogeneous families. Consequently, a stretching coordinate cannot replace a torsion or a ring coordinate during rank reduction.

The agreement with reference-quality refinements is therefore obtained after removing the molecule-specific model-design step, not after tuning it.

The three anhydrides were selected as a homologous literature-data series with a common chemical motif but progressively different experimental completeness, and therefore test different parts of the model layer. Maleic anhydride is the fully determined case: the published work of Vogt, Demaison, and Rudolph<sup>22</sup> reports the parent geometry, ground-state rotational constants, rovibrational corrections, and semiexperimental constants for a sufficiently rich isotopic set. Phthalic anhydride tests mixed estimation: Belyakov et al.<sup>23</sup> supplement the rotational data with weighted  $r_e^{\text{BO}}$  predicates. Succinic anhydride tests the opposite limit: Jahn et al.<sup>24</sup> provide only heavy-atom substitutions, so the hydrogen frame is not experimentally resolved and must be represented by explicit model relations.

This series also makes the input claim concrete. Once

$B_0$  constants and rovibrational corrections  $\Delta B_{\text{vib}}$  are available, the user transcribes the parent Cartesian geometry, the isotope substitutions, and the tabulated corrections. The program then perceives the topology and symmetry, constructs the non-redundant coordinate model, and the model layer applies any necessary constraints, predicates, or parameter classes as explicit statistical relations.

For maleic anhydride no auxiliary relation is introduced and 21 non-redundant symmetry coordinates are generated, of which the eight  $A_1$  coordinates are the active variables for the planar SE fit (Table III). The GIC and symmetry-Cartesian routes both give rank 8 and a rotational RMS residual of 0.036 MHz; the recovered bond lengths agree with the published  $r_e^{\text{SE}}$  values within  $1.2 \times 10^{-4}$  Å.

Table IV illustrates the type of chemically meaningful relations generated by the advisor. The user does not select active coordinates or construct a Z-matrix; instead, the program identifies unsupported structural directions and expresses them as primitive internal-coordinate relations, which are subsequently projected onto the algorithmically generated active space. The same representation also makes ring coordinates explicit: endocyclic bends are generated from primitive valence angles, while ring-puckering descriptors such as  $q$  and  $\phi$  are constructed from the endocyclic dihedral manifold rather than selected as molecule-specific torsions.

For phthalic anhydride, the rotational information alone can give a small rotational residual while leaving the structural model rank deficient. The published solution is therefore naturally expressed as primitive-

TABLE II. Decision burden removed by algorithmic model generation. Manual setup time is not reported because it depends on operator choices; the table therefore reports the reproducible decision burden. The final column counts molecule-specific coordinate or constraint edits made by the user in the reported runs.

| Case       | Manual workflow would decide                                                   | Automated model decisions | Generated model                     | Constraints/classes               | Result         | Manual edits |
|------------|--------------------------------------------------------------------------------|---------------------------|-------------------------------------|-----------------------------------|----------------|--------------|
| Azulene    | fused-ring coordinates; symmetry; active variables; planar pair                | 4                         | 48 GICs; 17 symmetric; 10 variables | 8 primitive constraints           | 0.073 MHz RMS  | 0            |
| Norcamphor | bridged coordinates; active variables; H-frame constraints; covariance targets | 4                         | 48 GICs; 48 active; 18 variables    | 30 H-frame constraints            | 0.071 MHz RMS  | 0            |
| Glycine I  | symmetry; unsupported H-frame directions; active variables                     | 3                         | 24 GICs; 15 $A'$ ; 11 variables     | 4 QC constraints                  | 0.0269 MHz RMS | 0            |
| Glycine II | low-symmetry active space; coupled descriptors; unsupported directions         | 3                         | 24 GICs; 19 variables               | 5 QC constraints; coupled classes | 0.249 MHz RMS  | 0            |

TABLE III. Algorithmically generated symmetry-coordinate model for maleic anhydride from literature data. Only the totally symmetric  $A_1$  block is active in the  $C_{2v}$  semiexperimental refinement.

| Symmetry block | Coordinates | Role in the SE model             |
|----------------|-------------|----------------------------------|
| $A_1$          | 8           | Active stretches/bends; rank 8   |
| $A_2$          | 3           | Torsion/out-of-plane diagnostics |
| $B_1$          | 3           | Torsion/out-of-plane diagnostics |
| $B_2$          | 7           | Stretch/bend diagnostics         |
| Total          | 21          | Non-redundant GIC space          |

coordinate predicates with conservative uncertainties of  $0.002 \text{ \AA}$  for bond lengths and  $0.2^\circ$  for angles. With these predicates, the fit has rank 14, condition number 4.05, and a rotational RMS residual of 0.50 MHz. It recovers the published  $r_e^{\text{SE}}$  structure within  $0.0015 \text{ \AA}$  for the tabulated bond lengths and within  $0.09^\circ$  for the tabulated angles.

Succinic anhydride completes the series because only heavy atoms are isotopically substituted. The heavy-atom framework is determined, but the hydrogen positions are not independent spectroscopic observables. In this case a local hydrogen-frame constraint gives a stable fit with rank 5 and a rotational RMS residual of 0.035 MHz. Weighted C–H predicates are still useful diagnostically, but as soft observations they do not remove all unobserved hydrogen-frame directions; without an additional topological or constraint relation the optimizer can enter geometrically equivalent but chemically unacceptable valleys. The comparison makes the distinction operational: constraints remove unsupported directions, predicates add observations with declared uncertainty, and parameter classes tie equivalent corrections.

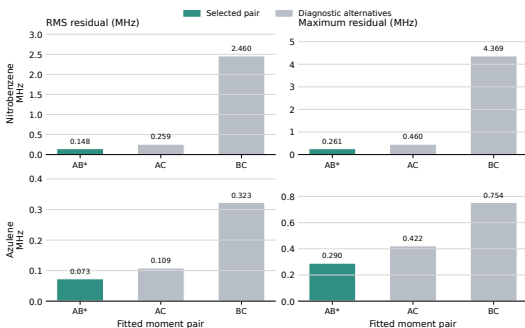


FIG. 4. Planar-component selection for nitrobenzene and azulene: the selected  $AB$  pair is highlighted and minimizes the complete post-fit diagnostic residual while the omitted component remains an external check. Detailed values are reported in the Supporting Information.

For planar molecules the model layer selects the best pair of rotational constants to be fitted, and retains the third rotational constant as an external check computed from the final Cartesian geometry. Figure 4 shows that the  $AB$  pair gives the smallest complete diagnostic residual for both nitrobenzene and azulene.

The condition numbers in the benchmark table should be interpreted as diagnostics of the inverse problem, not as residual-quality measures. Nitrobenzene is the clearest example. It fits only the planar  $AB$  component pair from nine isotopologues while retaining 13 active variables and eight primitive constraints; several deuterated substitutions therefore carry substantial leverage on directions that are weakly separated by the moments of inertia. The result is a high condition number ( $2.75 \times 10^4$ ) even though the final rotational RMS residual is only 0.148 MHz. Azulene, by contrast, has a larger fused-ring coordinate space but the algorithmic reduction leaves only 10 effective variables, and the additional isotopic coverage and primitive constraints distribute the spectroscopic information more evenly. Its condition number is therefore lower ( $9.57 \times 10^2$ ) despite the more complex topology. The same reading applies to the other benchmarks: high condition numbers flag directions whose uncertainties or influence diagnostics must be inspected, whereas acceptable residuals alone do not prove that the structural model is statistically well supported.

Table VI gives a direct sensitivity check. Maleic anhydride is stable under exact leave-one-isotopologue refits: the worst omitted-isotopologue RMS is 0.086 MHz and the largest Cartesian shift is  $5 \times 10^{-4} \text{ \AA}$ . Nitrobenzene remains predictive for most omissions, but the deuterated isotopologues expose the expected weak directions: the worst omitted RMS rises to 0.388 MHz and one refit reaches a condition number of  $3.7 \times 10^5$ . This is precisely the regime in which the algorithmic model is useful to an operator: it does not hide the problem behind a small training residual, but reports the rank, condition number, influence, and leave-one-out diagnostics needed to decide whether additional data or stronger model rela-

TABLE IV. Chemically readable primitive-coordinate relations generated by the model layer. The first four entries are representative reduced-dimensionality constraints for glycine; the last two show the corresponding form of generated ring coordinates. Atom labels are used for readability.

| Primitive relation  | Generated relation               | Role                                        |
|---------------------|----------------------------------|---------------------------------------------|
| O–H bond            | $ROH = R(O,H)$                   | fixes unsupported O–H distance              |
| C–O–H angle         | $AOH = A(C,O,H)$                 | preserves hydroxyl orientation              |
| H–N–H angle         | $HNH = A(H,N,H')$                | constrains amino frame                      |
| NH <sub>2</sub> wag | $WNH_2 = OOP(H,N,H',C)$          | removes unsupported wagging direction       |
| Endocyclic bend     | $RB = \sum c(i) A(i,j,k)$        | defines ring valence coordinate             |
| Ring puckering      | $q, \phi = f(D\text{-ring set})$ | reports puckering from endocyclic dihedrals |

TABLE V. Effect of auxiliary model relations for phthalic anhydride. The predicate model uses the conservative  $r_e^{BO}$  uncertainties reported by Belyakov et al. and the same literature rotational data. Structural errors are maximum absolute deviations from the tabulated  $r_e^{SE}$  bond lengths or angles.

| Auxiliary relation    | Rank | Cond.    | RMS    | Error           |
|-----------------------|------|----------|--------|-----------------|
| None                  | 13   | $\infty$ | 0.0065 | 3.13°           |
| Frozen H frames       | 9    | 1056     | 0.0067 | 0.19°           |
| $r_e^{BO}$ predicates | 14   | 4.05     | 0.50   | 0.0015 Å; 0.09° |

TABLE VI. Leave-one-isotopologue sensitivity diagnostics for representative refinements. The table reports exact refits in which one isotopologue is omitted, followed by prediction of the omitted rotational constants.

| System           | Refits | Worst omitted RMS<br>MHz | Omitted | Max Cartesian shift<br>Å | Cond. range                         |
|------------------|--------|--------------------------|---------|--------------------------|-------------------------------------|
| Maleic anhydride | 10     | 0.086                    | 13C2d2  | 0.0005                   | $6.1 \times 10^2 - 8.0 \times 10^2$ |
| Nitrobenzene     | 9      | 0.388                    | 3D      | 0.0840                   | $2.6 \times 10^4 - 3.7 \times 10^5$ |

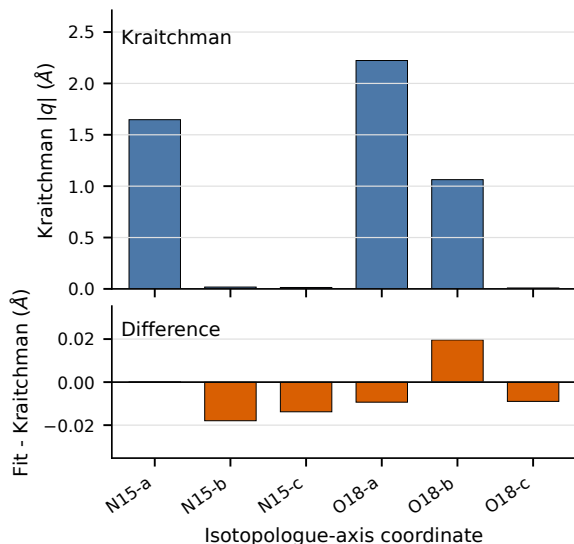


FIG. 5. Kraitchman diagnostic for the nitrobenzene planar benchmark. The upper panel reports absolute substitution coordinates from the Kraitchman analysis. The lower panel reports signed differences between the final SE coordinates and the Kraitchman values on an expanded scale. Numerical values are reported in the Supporting Information.

tions are required.

The next two diagnostics compare the fitted structures with independent structural information without

changing the least-squares objective. The nitrobenzene Kraitchman comparison in Figure 5 confirms that the fitted Cartesian structure follows the substitution-coordinate pattern, while the fit itself remains based on moments of inertia. This is the intended use of Kraitchman information here: it validates the SE+fit geometry but does not replace the statistically coupled refinement.

The Gly-I structural comparison in Figure 6 shows the scale of the semiexperimental correction when the starting QC geometry already reproduces the rotational data well. Since the reduced-dimensionality constraints are generated from the QC reference structure, the small SE–QC differences provide an a posteriori validation of the generated constraint model, not merely a check that the starting geometry was good.

Detailed atom numbering, rotational constants, topological parameters, Kraitchman diagnostics, reference-geometry comparisons, residuals, conditioning, and covariance diagnostics are reported in the Supporting Information. The main text retains the information needed to assess model formulation: rank, dimensionality, symmetry, constraint projection, planar-component choice, residual scale, and minimum checks.

#### A. Generated Reduced-Dimensionality Models

Incomplete isotopic coverage can leave some structural directions unsupported. The advisor therefore treats reduced dimensionality as part of the model-generation step. It preserves the highest compatible symmetry, identifies local coordinates that cannot be determined independently, and replaces them by primitive-coordinate constraints, predicates, or parameter classes.

Table VII shows that the reduced models are generated from isotope coverage and chemical equivalence rather than imported as a manual parameter subset. In glycine I, glycine II, and norcamphor the available substitutions

Glycine I fitted structural parameters and shifts from the PCS4 reference

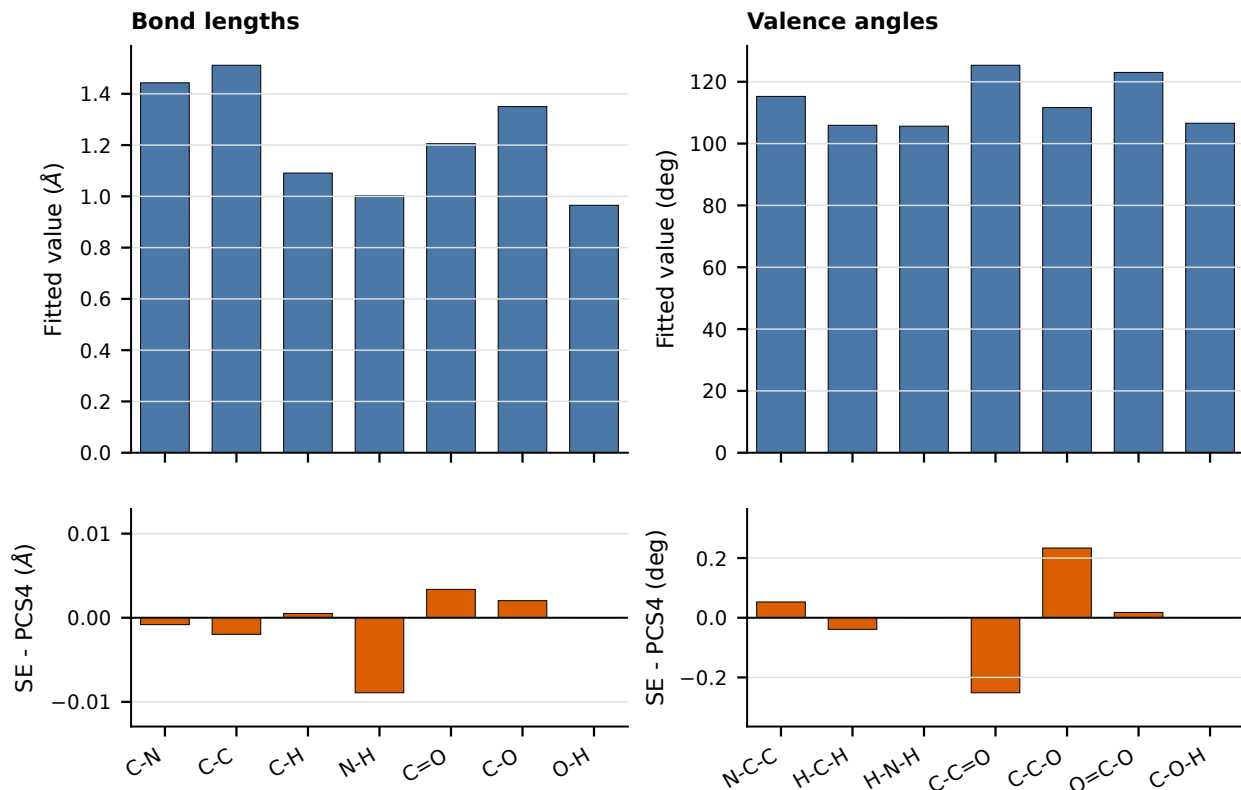


FIG. 6. Glycine I fitted structural parameters and shifts from the QC reference geometry. The upper panels report final SE bond lengths and valence angles; the lower panels report signed SE–QC differences on expanded scales.

TABLE VII. Autonomous structural-model decisions made by the advisor in the reduced-dimensionality benchmarks. “Active variables” are the independent least-squares coordinates after symmetry selection and projection of hard primitive-coordinate constraints.

| System     | GIC space        | Variables | Generated constraints         | Classes              |
|------------|------------------|-----------|-------------------------------|----------------------|
| Glycine I  | 24 GICs; 15 $A'$ | 11        | 4 QC functional constraints   | none                 |
| Glycine II | 24 GICs; $C_1$   | 19        | 5 QC functional constraints   | coupled descriptors  |
| Norcamphor | 48 GICs; $C_1$   | 18        | 30 hydrogen-frame constraints | constrained H frames |

do not support all hydrogen-related directions independently; the advisor therefore converts the unsupported directions into explicit primitive-coordinate relations derived from the reference structure.

## B. Glycine as an End-to-End Model-Generation Test

The two glycine conformers test the same model-generation layer starting from QC geometries when hydrogen-related directions are only partly supported.

For glycine I, the parent,  $^{13}\text{C}(\text{carboxyl})$ ,  $^{13}\text{C}(\text{methylene})$ ,  $\text{CD}_2$ , and  $^{15}\text{N}$  isotopologues are fitted using the rotational constants and weights of Godfrey and Brown<sup>25</sup> and the zero-point corrections of Kasalová et al.<sup>26</sup> The QC reference already reproduces

the 15 vibrationally corrected constants with an RMS relative error of 0.0258% and a maximum error of 0.0363%. The advisor detects the  $C_s$  geometry and incomplete hydrogen information and generates four constraints: O–H distance, C–O–H angle, H–N–H angle, and  $\text{NH}_2$  wag. Projection leaves 11 independent variables from 15  $A'$  GICs, and the final fit gives a complete rotational RMS residual of 0.0269 MHz with a positive optimized-space Hessian.

The Gly-I comparison is intentionally stringent: the refinement starts from an already spectroscopically consistent QC structure rather than correcting a poor starting point. Since the reduced-dimensionality constraints are generated from the QC reference structure, the small SE–QC differences provide an a posteriori validation of the generated constraint model. The largest heavy-atom

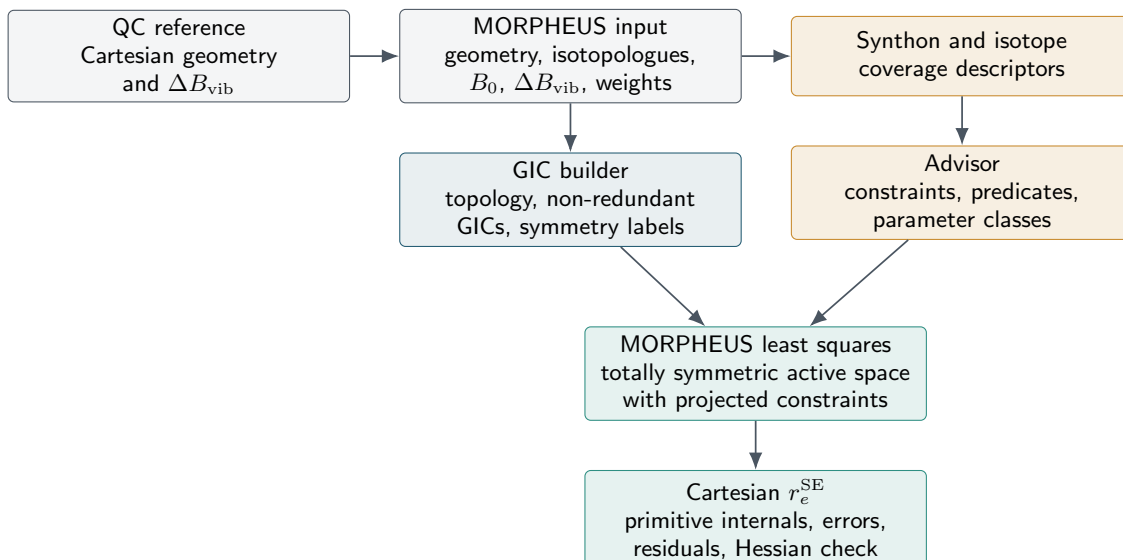


FIG. 7. End-to-end glycine refinement starting from QC Cartesian reference geometries and vibrational corrections. The calculation uses the common job/xyzin container, chemical descriptors, and isotope coverage to generate the active coordinate and constraint model without manually selected fit variables.

TABLE VIII. Glycine I rotational-constant diagnostics for the QC reference geometry and the final MORPHEUS semiexperimental geometry. Errors are computed against vibrationally corrected experimental constants for the five-isotopologue data set.

| Geometry     | RMS/MHz | Max/MHz | RMS/%  | Max/%  |
|--------------|---------|---------|--------|--------|
| QC reference | 1.7414  | 3.4072  | 0.0258 | 0.0363 |
| SE fit       | 0.0269  | 0.0571  | 0.0005 | 0.0011 |

TABLE IX. Parent glycine I rotational constants from QC and MORPHEUS/SE geometries. Corrected experimental and calculated values are in MHz.

| Component | Exp. $B_e$ | QC         | QC-Exp. | SE         | SE-Exp. |
|-----------|------------|------------|---------|------------|---------|
| A         | 10420.1910 | 10422.9024 | 2.7114  | 10420.2481 | 0.0571  |
| B         | 3907.4785  | 3907.5931  | 0.1146  | 3907.4786  | 0.0001  |
| C         | 2934.6809  | 2935.7175  | 1.0366  | 2934.6808  | -0.0001 |

bond and valence-angle shifts relative to the final semiexperimental structure are 0.0034 Å and 0.25°, respectively, both within the scale expected for state-of-the-art QC structures and vibrational corrections.<sup>2,11</sup>

Glycine II is the non-symmetric reduced-dimensionality case. The available isotopologues do not determine a stable full structure, so the QC reference is used to generate coupled primitive-coordinate constraints: C–H and N–H differences, two methylene angular combinations, and an average H–N–C–C torsion. These constraints are then projected onto the 24-dimensional  $C_1$  GIC space.

Table X shows that QC-derived targets produce the same rank and essentially the same residual scale as the published Kasalová-type constraints. The final QC-constrained refinement has 19 effective variables, a con-

TABLE X. Glycine II constraint-source comparison. Both calculations used GIC coordinates, moments of inertia as observables, and the same 10 corrected Glycine II rotational data sets. Rotational residuals are reported over the diagnostic A, B, and C constants in MHz. The last column is the mean-square convention used for comparison with the published glycine refinement, not a second RMS value.

| Constraint target source | Rank | RMS    | Max    | $10^3 \langle \Delta B^2 \rangle$ |
|--------------------------|------|--------|--------|-----------------------------------|
| Kasalová-type values     | 19   | 0.2344 | 0.8100 | 54.94                             |
| QC-derived values        | 19   | 0.2491 | 0.6707 | 62.06                             |

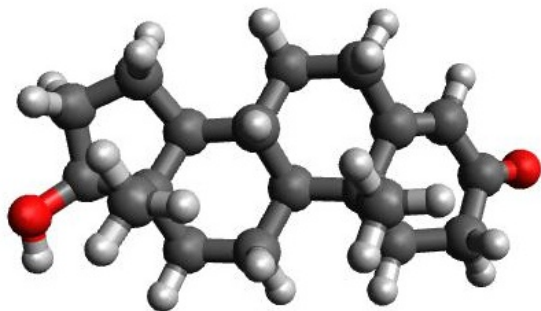
dition number of  $1.05 \times 10^4$ , and a complete rotational RMS residual of 0.2491 MHz. The largest uncertainties remain in the amino-hydrogen frame, as expected from the substitution pattern.

### C. Testosterone: Large-Molecule Scale Demonstration

The validation benchmarks above were selected because they represent available experimental gold standards while spanning acyclic, cyclic, fused, bridged, planar, homologous, and reduced-dimensionality cases. The model-generation layer is not limited to these sizes. As a scale check, the framework was also applied to testosterone, using the published experimental ground-state rotational constants and vibrational corrections for the observed T1-H2<sup>−</sup>-gp conformer.<sup>27</sup> This is a useful large molecule test because this level of treatment is rarely available for systems of this size: an accurate QC equilibrium geometry, accelerated  $\Delta B_{\text{vib}}$  corrections, and algorithmic model generation are combined in one repro-

ducible protocol.

The unrefined  $QC + \Delta B_{\text{vib}}$  residuals with respect to the corrected experimental constants are 4.638, 0.674, and 0.563 MHz for  $A$ ,  $B$ , and  $C$ , respectively. After a 141-dimensional non-redundant space is built, the class layer applies a single graph-defined C–C skeletal stretch class. Operationally, non-skeletal stretch combinations and the angular, torsional, and out-of-plane families are held fixed for this parent-only diagnostic, while the retained C–C stretch block is replaced by one shared variable after rank reduction. The resulting refinement improves the residuals to  $-0.027$ ,  $0.633$ , and  $0.291$  MHz, reducing the RMS rotational residual from 2.73 to 0.40 MHz. This test is deliberately not presented as an isotope-rich structural validation: it demonstrates that a large-molecule QC geometry, vibrational corrections, non-redundant coordinate generation, and class-constrained refinement can be chained for a 49-atom fused-ring molecule, where sufficiently rich isotopic substitution data are not presently available for a full validation of the fit engine.



Testosterone large-molecule scale demonstration.

Residuals are relative to the corrected experimental equilibrium constants derived from the published  $B_0$  and  $\Delta B_{\text{vib}}$  values.

| Model                        | Variables | $\Delta A$ | $\Delta B$ | $\Delta C$ |
|------------------------------|-----------|------------|------------|------------|
| $QC + \Delta B_{\text{vib}}$ | 0         | 4.638      | 0.674      | 0.563      |
| Class-constrained fit        | 1         | -0.027     | 0.633      | 0.291      |

## VI. CONCLUSIONS

The central contribution of this work is a general framework for constructing constrained refinement models, demonstrated here through semiexperimental structure refinement. Existing refinement methods generally assume that the coordinate model has already been supplied; here topology perception, symmetry assignment, homogeneous rank pruning, constraint projection, least-squares solution, and covariance propagation are part of one calculation rather than separate user decisions.

The benchmarks show that the resulting models remain graph-defined, symmetry-consistent, rank-consistent, and non-redundant for acyclic, cyclic, planar, fused, bridged, homologous anhydride, glycine, and large fused-ring cases. The program selects the active symmetry space, removes redundant coordinates, chooses the

independent moment components for planar molecules, generates constraints when isotopic coverage is incomplete, and propagates covariance to the reported structural parameters. Where both active coordinate models are applicable, GIC and symmetry-Cartesian refinements give the same effective dimensions and rotational residuals.

Electronic-structure information is used only where the experiment does not fully determine the structure. For Gly-I the QC reference already lies within the practical rotational-accuracy window, so the fit provides a statistically consistent semiexperimental correction and uncertainty estimate. For Gly-II and norcamphor, the advisor converts reference information and isotope coverage into explicit reduced-dimensionality constraints. The reference therefore supplies constraints for unsupported directions, while the spectroscopic data determine all supported directions.

This is not simply a convenient input generator. The final outputs remain ordinary chemical objects: Cartesian structures, bond lengths, valence angles, dihedrals, ring descriptors, propagated uncertainties, residuals, and diagnostics. What changes is that the path from topology and data to those objects is explicit, reproducible, and rank-audited, even for molecular graphs in which manual coordinate formulation is often the limiting step.

The testosterone scale check extends this logic to the current upper end of practical rotational-spectroscopic modeling. It does not claim a complete semiexperimental structure from a parent-only data set; instead, it shows that an accurate large-molecule QC geometry, accelerated  $\Delta B_{\text{vib}}$  corrections, algorithmic non-redundant GIC generation, and class-constrained refinement can be chained for a 49-atom fused-ring molecule in a reproducible calculation. This closes the loop from accurate large-molecule QC structures to algorithmically generated spectroscopic refinement models. The framework therefore performs two coupled tasks simultaneously: it constructs the refinement model algorithmically and refines the initial equilibrium geometry toward the semiexperimental solution. Model generation and structural refinement are integrated into a single workflow rather than being treated as separate steps. The essential shift is from expert-dependent model formulation to reproducible algorithmic model generation. The algorithm replaces expert-dependent model formulation rather than expert chemical judgement. Although motivated by semiexperimental equilibrium structures, the underlying graph-driven model-construction strategy is applicable to any constrained refinement problem requiring simultaneous treatment of topology, symmetry, rank deficiency, and external structural information.

## SUPPORTING INFORMATION AVAILABLE

Atom-numbered structures for the benchmark and companion systems; complete rotational-constant com-

parisons for glycine I, glycine II, cyclopentadiene, nitrobenzene, and azulene; Kraitchman, reference-fit, and planar-component diagnostics; additional diagnostic tables for glycolaldehyde and norcamphor; final Cartesian coordinates, propagated topological parameters, residuals, and archived MORPHEUS output files.

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